Drawing Lines in U.S. Monetary and Fiscal history

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Abstract

A sequel to Sargent [2011] on the trade-off between price-level stability and Ramsey-optimal tax smoothing, this essay illustrates the practical delicacy of the institutional choice between fiscal dominance and monetary dominance by cataloguing celebrated reversals. Milton Friedman crossed the divide between his 1948 proposal for coordinating monetary and fiscal policy and his 1960 program for a fixed money-growth rule—and Paul Volcker's short-lived operationalization of that rule in 1979–82 demonstrated that sustaining it requires an institutional environment that the American financial system could not maintain. Henry Clay and James Madison each opposed the First Bank of the United States on constitutional and republican grounds, then became architects of the Second Bank once the War of 1812 had exposed the fiscal cost of having no national monetary institution. Daniel Webster voted against the charter of the Second Bank in 1816 but within three years had become its foremost legal champion. Albert Gallatin entered Congress as a Jeffersonian critic of Hamilton's financial system and became, as Secretary of the Treasury, the Bank's strongest executive-branch defender. Most dramatically, Salmon P. Chase spent a career as a hard-money Jacksonian Democrat, engineered the Legal Tender Acts as Secretary of the Treasury in the Civil War, and then, as Chief Justice, wrote the opinion in *Hepburn v. Griswold* (1870) declaring his own Acts unconstitutional. The essay then extends the American nineteenth-century record with three twentieth-century parallels. John Maynard Keynes traveled the opposite path from Friedman: from sophisticated defender of the gold exchange standard in 1913 to author of the most celebrated indictment of gold discipline in the English language, driven by his reading of the deflationary costs that the restored gold parity imposed on post-war Britain. Marriner

Eccles championed the Federal Reserve's wartime bond-price peg—making the Fed a fiscal instrument of the Treasury—and then, demoted by Truman for his efforts, became the leading institutional advocate of the Treasury-Fed Accord that restored monetary independence in 1951. Richard Nixon campaigned against Keynesian fiscal expansion and then, on a single Sunday evening in August 1971, closed the gold window, imposed wage-price controls, and declared himself "a Keynesian in economics." The essay also turns from individual reversals to institutional ones: the half-century redrawing of the line between *money* and *credit*, from the bond-backed currency of the National Banking Acts that Chase championed to the real-bills provisions of the Federal Reserve Act of 1913; and, as a British parallel, the story of Robert Peel's Bank Charter Act of 1844 and its three suspensions—a statutory line bent, not repealed, by the logic of acute financial crisis, and the occasion for Bagehot's inversion of Currency School orthodoxy. These reversals are not signs of weakness or inconsistency; they are evidence of how contextually sensitive, and how consequential, the institutional lines really are. They also reveal a structural pattern: reversals cluster at moments of systemic stress—wars, panics, and depressions—when the costs of the existing institutional arrangement suddenly become undeniable to multiple actors at once.

1 Introduction

Where to draw the institutional boundaries between monetary and fiscal authorities is an unsettling and consequently unsettled issue. Sargent [2011] argued that one source of difficulty is that arrangements that support price-level stability tend to foreclose the Ramsey-optimal tax smoothing that Robert E. Lucas and Stokey [1983] showed requires using money creation as a flexible fiscal instrument, while arrangements that enable efficient public finance tend to inject fiscal pressures into monetary policy and thereby risk inflation. Some of the most able statesmen and economists in American and British history changed their minds, sometimes more than once, and sometimes in ways that placed them in direct tension with their earlier selves. This essay examines six individual reversals in American history—Friedman, Clay, Webster, Gallatin, Madison, Chase—and three in the twentieth century that exhibit a similar same pattern—Keynes, Eccles, Nixon—and then turns to two institutional reversals that unfolded over decades: Congress's stepwise redrawing of the line be-

tween money and credit, from the bond-backed National Bank Notes of the 1860s to the real-bills-backed Federal Reserve notes of 1913; and, as a British parallel, Robert Peel's Bank Charter Act of 1844, whose statutory line was tested in three successive crises and bent without ever being formally repealed.

Milton Friedman, who serves as the primary example in Sargent [2011], moved from a 1948 proposal that coordinated money creation tightly with fiscal deficits—an arrangement that prioritizes efficient stabilization but makes monetary policy subordinate to budget outcomes—to the 1960 program for a fixed k-percent money-growth rule that insulates money entirely from the budget, forfeiting tax-smoothing efficiency in the name of price-level stability. The subsequent history of Friedman's rule adds a further chapter: Paul Volcker's Federal Reserve operationalized it in October 1979, only to abandon it by 1982 when financial deregulation destabilized the very monetary aggregates on which the rule depended.

Henry Clay and James Madison present a complementary pattern at an earlier moment in American history. Both opposed the First Bank of the United States—Clay as a young Jeffersonian congressman on republican and states'-rights grounds, Madison as leading constitutional theorist of the founding generation. Financial challenges brought with the War of 1812, fought without a national bank capable of mobilizing government credit, influenced both of them. Two other conversions illuminate the same period: Daniel Webster, a New England Federalist who voted against the Second Bank's charter in 1816, became its most prominent legal and political defender within three years; and Albert Gallatin, the Jeffersonian firebrand who criticized Hamilton's Treasury from the House floor in the 1790s, became the Bank's strongest advocate when executive responsibility as Secretary of the Treasury convinced him of its fiscal indispensability. Clay would make a national bank a centerpiece of his "American System"; Madison would sign the bill creating the Second Bank in 1816, recognizing that long legislative practice had reframed his constitutional concerns from 25 years earlier.

Perhaps the most dramatic individual reversal belongs to Salmon P. Chase. A hard-money Jacksonian Democrat whose early career was defined by opposition to paper currency and to the mixing of public and private credit, Chase became Abraham Lincoln's Secretary of the Treasury in 1861 and proceeded to design and defend the Legal Tender Acts of 1862 and 1863, which made inconvertible paper notes ($450

million in "greenbacks") legal tender for all debts. Appointed Chief Justice in 1864, Chase spent much of the rest of his career trying to undo his own handiwork. In *Hepburn v. Griswold* (1870) he wrote the majority opinion holding the Legal Tender Acts unconstitutional as applied to debts contracted before their passage. Grant's appointment of two new justices reversed that decision in *Knox v. Lee* (1871). Chase dissented.

The twentieth-century parallels extend the pattern in instructive directions. John Maynard Keynes moved from sophisticated defender of the gold exchange standard to its most celebrated opponent, driven by his reading of the deflationary costs that gold imposed on post-war Britain. His trajectory is the mirror image of Friedman's: both men were updating the same trade-off in response to the same historical events, but each concluded that the other's preferred failure mode was the greater danger. Marriner Eccles championed the wartime bond-price peg that made the Federal Reserve a fiscal instrument of the Treasury, then reversed course and became the leading institutional advocate for monetary independence—at the cost of his chairmanship. Richard Nixon, campaigning on fiscal conservatism and sound money, closed the gold window and embraced Keynesian stimulus in a single Sunday announcement in August 1971.

My theme is not that these figures were confused or hypocritical. Each reversal was accompanied by sustained reasoning and, usually, by an explicit account of why changed circumstances or changed understanding warranted a different institutional line. What the reversals collectively demonstrate is that the stability-versus-efficiency trade-off identified in Sargent [2011] is not merely a theoretical curiosity; it involves a real live political and constitutional choices that intelligent people answer differently in different contexts, and sometimes answer differently over the course of a single lifetime.

One further structural pattern deserves notice from the outset. The reversals documented here cluster at moments of systemic stress. The War of 1812 converted Clay and Madison simultaneously and vindicated Gallatin's earlier administrative conclusions. The Civil War broke Chase and funded the bond-backed currency system. The Great War and its deflationary aftermath drove Keynes's evolution. World War II created the circumstances that confronted Eccles. Nixon's 1971 reversal was in response to fiscal consequences of Vietnam War spending. This clustering suggests

that the decisive "forcing function" is not gradual evidence-accumulation alone, but crisis-induced surprises or revelations: costs of existing institutional arrangement that were tolerable in a steady fiscal environment suddenly become undeniable to multiple actors.

2 The Framework Recalled

Sargent [2011] builds on the Sargent and Wallace [1981] "unpleasant monetarist arithmetic" framework. Let the government's flow budget constraint be

$$b_t + \tau_t + \mu_t m_{t-1} = g_t + (1 + r_{t-1})\, b_{t-1}, \tag{1}$$

where b_t is real debt, τ_t real tax revenues, g_t real spending, r_{t-1} the real interest rate on debt, and $\mu_t = (M_t/M_{t-1} - 1)$ the money-growth rate applied to a real money base m_{t-1}. Seigniorage $\mu_t m_{t-1}$ enters as a source of revenue that competes with ordinary taxation.

Two polar institutional arrangements are possible:

Monetary dominance. The monetary authority commits credibly to a path for $\{M_t\}$ (or equivalently the price level), and the fiscal authority must choose $\{\tau_t,\, b_t,\, g_t\}$ to satisfy (1) without altering that path. Fiscal policy is the "residual claimant"; seigniorage is fixed exogenously by the monetary rule. Price-level stability is guaranteed by the commitment, but tax-smoothing efficiency is constrained: the fiscal authority cannot use seigniorage as a flexible instrument to smooth distortionary taxes across states and dates.

Fiscal dominance. The fiscal authority sets $\{\tau_t,\, g_t,\, b_t\}$ autonomously and the monetary authority passively accommodates whatever seigniorage is required to clear (1). Ramsey-optimal tax smoothing is now achievable: if shocks to g_t are sufficiently persistent, Robert E. Lucas and Stokey [1983] show that optimality calls for using money creation alongside other taxes. But fiscal dominance makes the price level a fiscal consequence; a deterioration in fiscal discipline is monetized and price-level increases follows.

The core result of Sargent [2011] is that neither arrangement Pareto dominates the other. Monetary dominance trades efficiency for stability; fiscal dominance trades

stability for efficiency. The social ranking depends on parameterization and context. This is the sense in which a "line" is not obvious: reasonable people, reasoning carefully, can end up on different sides depending on their assessment of the relevant fiscal and monetary environment. The historical reversals documented below are vivid illustrations.

3 Milton Friedman: From Fiscal Coordination to Monetary Rules

Friedman's 1948 paper in the *American Economic Review*, "A Monetary and Fiscal Framework for Economic Stability" [Friedman, 1948], proposed an institutional architecture in which monetary policy was explicitly subordinated to fiscal outcomes. The proposal rested on three pillars: (1) a requirement that commercial banks hold 100 percent reserves against deposits, so that money creation was entirely a government function; (2) automatic fiscal stabilizers (progressive taxation and transfer payments) that would cause the budget to move into surplus during booms and deficit during recessions without discretionary action; and (3) a rule specifying that any deficit be financed exclusively by issuing money, while any surplus be used to retire money. Under this system, the supply of money was mechanically linked to the fiscal balance: the government literally printed money to finance shortfalls and destroyed money when revenues exceeded expenditures.

This is a case of programmatic fiscal dominance. The monetary instrument—the money supply—is not controlled by an independent central bank pursuing a price-level target; it is set residually to satisfy a fiscal accounting identity. In the terminology of Sargent [2011], the monetary authority has been dissolved into the fiscal authority. Friedman embraced this design in 1948 because he believed that the economic cycle was predominantly driven by changes in aggregate demand, that discretionary monetary and fiscal management were dangerously unreliable, and that the automatic, formula-driven coupling of money creation to fiscal outcomes would deliver macroeconomic stabilization without giving policymakers enough discretion to do harm. His touchstone at this stage was the *coordination* of the two policy arms, not their separation.

By the late 1950s Friedman had moved to an opposite point of view. His Fordham

University lectures, published as *A Program for Monetary Stability* [Friedman, 1960], advocated for a fixed k-percent annual growth rate in a well-defined monetary aggregate, entirely independent of fiscal conditions. The Federal Reserve should expand the money supply at a constant rate—Friedman suggested roughly 3 to 5 percent per year—regardless of whether the budget was in surplus or deficit. This is a textbook case of monetary dominance: the monetary rule is the binding commitment, and the fiscal authority must accommodate it.

The intellectual distance between the two positions is large. Under the 1948 framework, a persistent fiscal deficit would automatically generate money creation and hence inflation; the 1960 framework explicitly blocks this channel. What changed?

Friedman identified several reasons [Friedman, 1960], Chapter 4. First, he had become convinced by the historical evidence, which he and Friedman and Schwartz [1963] were then assembling, that autonomous monetary shocks—changes in the money stock not driven by fiscal conditions—were the dominant source of macroeconomic fluctuations. The Great Depression was not a collapse of aggregate demand correctable by fiscal-monetary coordination; it was a catastrophic monetary contraction engineered by Federal Reserve mistakes. If monetary shocks were the main danger, the right institutional response was to bind the monetary authority, not to couple it to fiscal outcomes.

Second, Friedman had concluded that fiscal discipline in a democratic polity was essentially unenforceable. Governments that could finance deficits by money creation would reliably do so. The 1948 design assumed that the fiscal stabilizers would be genuinely automatic and symmetric: surpluses in good times would destroy money just as deficits in bad times created it. But the historical record suggested that democracies enlarged their fiscal commitments during downturns and failed to retire them during expansions. A design that coupled monetary policy to fiscal outcomes would therefore degenerate into a one-way ratchet toward inflation.

Third, and most analytically interesting from the perspective of Sargent [2011], Friedman had implicitly revised his assessment of the stability-efficiency trade-off. The 1948 framework prized efficiency (smooth automatic stabilization) and accepted the price-level risk entailed by fiscal coupling. The 1960 program prized stability (a committed monetary rule) and accepted the efficiency loss from foreclosing flexible

seigniorage. The change was not a logical correction of an earlier error; it was a shift in the relative weight placed on the two objectives, driven by an updated reading of what the American political economy could actually sustain.

Friedman's reversal is a clean illustration of the paper's main thesis: the optimal institutional line between fiscal and monetary authority depends on empirical and institutional features of the environment that are themselves changing and contestable.

3.1 Paul Volcker and the Test of Friedman's Rule

Friedman's k-percent rule was not merely an academic proposal; it was operationalized—once—at the Federal Reserve. On October 6, 1979, Paul Volcker announced what the Fed called the "New Operating Procedures": rather than targeting the federal funds rate, the Federal Open Market Committee would henceforth target non-borrowed reserves, in practice allowing interest rates to fluctuate freely while controlling the growth of monetary aggregates. The announcement was widely interpreted, including by Friedman himself, as the closest the American central bank had ever come to implementing the k-percent rule. Volcker had not adopted Friedman's specific prescription—he was targeting the path of money growth, not a fixed percentage—but the animating principle of the 1979 regime was identical to the 1960 proposal: bind the monetary instrument, accept short-term interest-rate volatility, and let the fiscal authority accommodate the monetary path rather than the reverse.

The monetarist experiment lasted until the summer of 1982—roughly thirty months. What ended it was not a failure of will on Volcker's part but an institutional problem that Friedman had identified in the abstract yet had not fully anticipated in practice. The financial deregulation of the early 1980s—the Depository Institutions Deregulation and Monetary Control Act of 1980, the proliferation of interest-bearing checking accounts (NOW accounts), and the growth of money-market mutual funds—fundamentally altered the demand function for the monetary aggregates that served as the rule's target. Velocity became unpredictable; the historical relationship between M1 growth and nominal income that had justified M1 as an intermediate target broke down. The aggregates were sending contradictory signals about the stance of policy.

By 1982, the recession induced by the disinflation was pushing unemployment toward 11 percent, the highest since the Great Depression. The collapse of Penn Square Bank

in July 1982 and Mexico's announcement in August 1982 that it could not service its sovereign debt threatened systemic financial instability. Volcker, without formal announcement, shifted back to interest-rate targeting.

His memoir [Volcker and Harper, 2018] reflects explicitly on the episode. Volcker acknowledged that the monetary aggregates had become unreliable due to financial innovation, but also maintained that the 1979–82 disinflation had accomplished its primary objective—breaking inflationary expectations—even at the cost of an acute recession. In the terms of Sargent [2011], the Volcker experiment confirms the same lesson that Friedman's own reversal had taught: the optimal institutional design depends on features of the financial environment that change—sometimes faster than the institutions designed to manage them. Friedman had argued for the k-percent rule because he believed the money-demand function was stable enough to make the monetary aggregate a reliable intermediate target. The deregulation of the 1980s undermined that stability, validating Friedman's rule in principle while rendering it unworkable in the specific institutional environment in which its single attempt at implementation occurred.

4 Henry Clay: From Republican Critic to Champion of the American System

Henry Clay entered Congress as a Kentucky senator in 1806, steeped in the Jeffersonian Republican tradition that regarded the First Bank of the United States as an unconstitutional concentration of financial power in private hands. When the First Bank's twenty-year charter expired in 1811 and recharter was debated, Clay—now a newly elected member of the House—was among those who voted against it. The recharter bill died narrowly: the Senate divided 17–17, and Vice President George Clinton cast the deciding vote against; the House defeated it 65 to 64. Clay's opposition was rooted in the Jeffersonian political economy that Andrew Jackson would later systematize: the Bank was a creature of Hamilton's Federalist designs, an engine of privilege and corruption, constitutionally suspect, and dangerous to the democratic republic.

Five years later, Clay was one of the most forceful advocates in Congress for creating the Second Bank of the United States. The Second Bank was chartered in April 1816

with a twenty-year term, capitalized at $35 million (more than three times the First Bank's capitalization), and endowed with explicit authority to issue notes that could serve as currency nationwide. Clay's support was enthusiastic and, more importantly, principled: he incorporated the national bank into his "American System," the integrated program of protective tariffs, federally financed internal improvements, and a national monetary institution that he argued was necessary for American industrial development, national security, and economic integration.

What happened between 1811 and 1816 was the War of 1812. In the absence of a national bank, the Madison administration found it nearly impossible to finance the war. State banks proliferated and issued paper with no common standard; the government was reduced to borrowing at punishing terms from private financiers; military operations were hampered by the Treasury's inability to transfer funds efficiently across the continent. In 1814, state banks outside New England suspended specie payments—an event that dramatized the instability of a purely decentralized monetary system. The government's own fiscal operations were threatened: Treasury notes circulated at steep discounts, and the Secretary of the Treasury reported that he could not reliably meet the government's payroll.

Clay's reversal is intelligible as a response to this evidence. Before the War of 1812, the costs of a national monetary institution—the constitutional objections, the concentration of financial privilege, the threat to state banking interests—loomed large relative to its benefits, which seemed speculative and ideological. After the war, the corresponding cost-benefit calculation ran the other way: the demonstrated fiscal cost of the absence of a national institution was concrete and severe, while the constitutional objections had been blunted by the practical success of the First Bank during its twenty years of operation. The line between efficient national finance and the dangers of centralized monetary power had moved, not because the underlying principle had changed, but because the evidence on the costs of each institutional arrangement had accumulated.

In the framework of Sargent [2011], the First Bank had been an instrument of fiscal-monetary coordination: it held government deposits, facilitated tax collection, managed government debt, and provided a measure of monetary discipline through its willingness to redeem its notes. The two decades after its expiration amounted to an unplanned experiment in the absence of such coordination. Clay read the experimen-

tal results and updated accordingly. That is not inconsistency; that is rationality.

The sequel to Clay's conversion is worth noting. When Andrew Jackson vetoed the recharter of the Second Bank in 1832 and destroyed it by withdrawing government deposits in 1833–34, Clay emerged as the leading opponent of Jackson's action. The Bank War became the defining issue of Clay's Whig Party. The man who had voted against the First Bank in 1811 became, by the 1830s, the foremost defender of the principle of a national monetary institution against Jacksonian hard-money democracy. The reversal had become the fixed point from which he did not thereafter move.

4.1 Daniel Webster: From Opposing the Charter to Championing the Bank

Daniel Webster's trajectory offers a complementary reversal from the opposite end of the partisan spectrum. Webster entered the House of Representatives in 1813 as a Federalist from New Hampshire—the party of Hamilton that had created the First Bank. Yet when the Second Bank bill came before the House in 1816, Webster voted against it. His objections were not Clay's earlier Jeffersonian ones about constitutionality or the concentration of privilege; Webster opposed the *specific terms* of the bill, arguing that it failed to require prompt resumption of specie payments and that the Bank's capital structure and relationship to the government were poorly designed. He wanted a sounder national bank, not no national bank at all.

Within three years Webster had become the Bank's foremost legal champion. He argued the Bank's case before the Supreme Court in *McCulloch v. Maryland* (1819), the decision that vindicated the constitutionality of a national bank under the Necessary and Proper Clause and became a cornerstone of federal implied-powers doctrine. By the 1820s, Webster was on standing retainer as the Bank's counsel, a relationship he maintained throughout Nicholas Biddle's presidency of the institution. The financial entanglement was not incidental: Webster borrowed from the Bank and was frank about the connection, writing to Biddle on at least one occasion to note that "my retainer has not been renewed or refreshed as usual" and to inquire whether the Bank wished the relationship to continue. During the Bank War of the 1830s, Webster stood beside Clay as one of the two most formidable Senate opponents of Jackson's

campaign to destroy the institution.

Webster's reversal is less ideological than Clay's—he moved from objecting to a specific legislative design to championing the institution as a legal and political cause, rather than crossing a deep philosophical divide—but it illustrates an additional mechanism of conversion. Where Clay was compelled by the evidence of fiscal catastrophe during the War of 1812, Webster was converted by the Bank's *operational success* in its early years and by his growing appreciation, as the leading commercial lawyer of New England, of what a well-managed national monetary institution could accomplish for interstate commerce and the stability of private credit. The institution that looked ill-designed on paper in 1816 proved indispensable in practice by 1819. In the framework of Sargent [2011], Webster updated his assessment not of the *principle* of fiscal-monetary coordination but of the *institutional capacity* to execute it competently—precisely the kind of empirical updating that makes the optimal location of the line a moving target.

5 James Madison: Constitutional Principle and the Doctrine of Liquidation

James Madison's trajectory runs parallel to Clay's and, in some respects, deeper, because Madison's original opposition to the First Bank was not merely political but philosophical—grounded in the most careful constitutional theory of his generation.

When Alexander Hamilton proposed the First Bank in January 1791, Madison led the House opposition. His speech of February 2, 1791 is one of the classic statements of strict constructionism. The Constitution granted Congress enumerated powers. The power to charter a corporation was not among them. The "necessary and proper" clause could not be interpreted to extend congressional power to any means that might be deemed convenient or useful; "necessary" had to mean genuinely indispensable to the execution of an enumerated power, and chartering a bank was not indispensable to anything the Constitution explicitly authorized. Madison also invoked what would later be called the principle of original understanding: the framers' intentions, as manifested in the ratification debates, gave no support for so broad an implied power.

Hamilton's counter-argument won: Washington accepted Hamilton's opinion and

signed the bill. But Madison's position was not merely a tactical loss; it established the intellectual framework that Jeffersonian Republicans would invoke for the next two decades, and that Jackson's Democrats would recycle in the Bank War of the 1830s.

5.1 Albert Gallatin: From Jeffersonian Critic to Executive-Branch Champion

Albert Gallatin's trajectory illustrates the same conversion from the vantage point of executive responsibility. Gallatin arrived in Congress in 1795 as a Jeffersonian Republican from western Pennsylvania, steeped in the agrarian suspicion of Hamilton's financial architecture. As the leading Republican voice on fiscal affairs in the House, Gallatin subjected Hamilton's Treasury operations—including the operations of the Bank—to relentless scrutiny, viewing them as instruments of Federalist political consolidation rather than as neutral fiscal machinery. His opposition was pragmatic rather than constitutional: where Madison had argued that chartering a bank exceeded Congress's enumerated powers, Gallatin objected that Hamilton's Bank served Federalist patronage networks more than it served the public interest.

When Jefferson appointed Gallatin Secretary of the Treasury in 1801, Gallatin brought these Jeffersonian convictions to the office that Hamilton had designed. Administrative experience changed his mind rapidly. The Bank proved to be not the engine of partisan privilege that Gallatin had attacked from the House floor but an indispensable instrument of day-to-day fiscal management: it held government deposits safely, transferred funds efficiently across the continent, facilitated customs collections at dispersed ports, and provided a stable medium for servicing the national debt. By the time the Bank's twenty-year charter approached expiration, Gallatin had become its strongest advocate within the administration. In a formal report to Congress in 1809, he urged recharter in terms that would have been unthinkable from the Jeffersonian firebrand of the 1790s, arguing that the Bank was "necessary" for the responsible management of the government's finances and warning that its dissolution would impose severe fiscal costs.

The warning went unheeded. The recharter bill died in 1811—by the narrowest of margins, as noted in the preceding section—and the fiscal catastrophe of the War

of 1812 vindicated Gallatin's prediction in every particular. Gallatin, who served as a peace commissioner at Ghent in 1814, watched from abroad as the Treasury struggled with exactly the problems he had foreseen: no reliable depository for government funds, no mechanism for interstate fiscal transfers, no institutional anchor for public credit. The experience confirmed what his years at the Treasury had already taught him: that the abstract Jeffersonian case against a national bank could not survive contact with the operational realities of governing a continental republic. Gallatin's later writings on banking and currency, notably his *Considerations on the Currency and Banking System of the United States* (1831), drew explicitly on his executive experience to defend the principle of a national monetary institution against both Jacksonian hard-money populism and the proliferating disorder of unregulated state banking.

Gallatin's conversion is significant for the argument of this essay because it was driven not by the shock of wartime emergency—which came later—but by the quieter accumulation of administrative evidence. The War of 1812 confirmed what Gallatin already knew; it did not cause his reversal. In the language of Sargent [2011], Gallatin's experience as Treasury Secretary updated his estimate of the *efficiency gains* from fiscal-monetary coordination: the institution that had appeared, from the opposition benches, to be a source of corruption and privilege turned out, from the Treasury desk, to be a source of fiscal efficiency that no combination of state banks and ad hoc arrangements could replicate.

5.2 Madison's Reversal

In 1815, near the end of the war and his presidency, Madison vetoed a bank bill—not on constitutional grounds, but because the proposed bank lacked features that would make it an effective fiscal instrument for the government. The veto message explicitly declined to reopen the constitutional question, which Madison said had been settled by congressional and executive practice over twenty-four years. The following year, in 1816, Madison signed the bill creating the Second Bank of the United States.

How did the author of the 1791 speech against bank constitutionality come to sign the bill creating a larger successor? Madison's answer was sophisticated and, to his critics, infuriatingly convenient. He had articulated it already in the 1815 veto message: while his own constitutional views had not changed, the Constitution's

meaning was not fixed solely by the original understanding of individual framers. It was also determined by the accumulated legislative, executive, and judicial practice of the republic. A course of action repeatedly approved by Congress, acquiesced in by successive presidents, and never invalidated by the courts thereby acquired constitutional legitimacy through what Madison called the process of "liquidation"—the settlement of constitutional meaning by institutional practice over time. By 1816, twenty-four years of the First Bank's operation had, in Madison's view, liquidated the constitutional question in favor of congressional authority.

The liquidation doctrine is philosophically consequential. It implies that the institutional line between permissible and impermissible concentration of fiscal-monetary power is not fixed by original text alone; it evolves through a common-law-like process of precedent and practice. Institutions that were once constitutionally contested can become constitutionally settled, and the initial objections correspondingly lose their force. In the terms of Sargent [2011], the boundary between stability-promoting and efficiency-promoting arrangements is not only analytically ambiguous; it is constitutionally labile, subject to reinterpretation as practice accumulates.

Madison's reversal also reflects the same lesson Clay drew from 1812: the War had revealed the fiscal consequences of the absence of a national monetary institution. A republic that could not finance its wars without submitting to ruinous private financiers was not well served by constitutional principles, however impeccably derived. The stability of the polity—its fiscal survival—was itself at stake; abstract monetary-constitutional purity had proved a luxury that the republic could not afford.

The most important contrast to Madison is his friend and rival Thomas Jefferson, who never accepted the liquidation doctrine and to the end of his life insisted that the Bank was unconstitutional. Jefferson's consistency and Madison's flexibility represent the two sides of what is today still a live debate: whether institutional practice can and should revise constitutional meaning, or whether original principle must constrain practice however inconvenient the results. That debate is, in the end, about who gets to move the lines—and whether the lines can legitimately be moved at all.

6 Salmon P. Chase: From Hard Money to Legal Tender to Constitutional Revision

The most internally dramatic of the reversals documented here belongs to Salmon P. Chase. His career traces a complete arc: from principled hard-money doctrine, through the coercion of wartime necessity, to a judicial crusade to undo his own executive work. Few episodes in American constitutional history have been dissected more searchingly than this one. The fullest contemporary anatomy is George Bancroft's *A Plea for the Constitution of the United States Wounded in the House of Its Guardians* [Bancroft, 1886], a remarkable document that repays study here.

6.1 A Witness of Singular Authority

Bancroft (1800–1891) was uniquely positioned to indict the Legal Tender Cases. As the foremost narrative historian of the United States—his ten-volume *History of the United States* (1834–74) was the monumental scholarly achievement of the century—he brought to monetary constitutionalism not merely the moralism of a hard-money partisan but the documentary authority of a man who had spent fifty years in the primary sources of the founding era. He had personally known James Madison and John Adams; had served as Secretary of the Navy under Polk and as Minister to Prussia and Germany under Johnson and Grant; and was 84 years old when, in the wake of *Juilliard v. Greenman* (1884), he sat down to write his *Plea*. The book is a comprehensive historical argument that the framers at Philadelphia had deliberately and by large majorities stripped from the federal government any power to make paper money legal tender for private debts—and that a sequence of Supreme Court decisions had betrayed that design. His central exhibit, running through every part of the work, is Salmon P. Chase.

Bancroft's constitutional argument rests on what happened in Philadelphia on August 16, 1787. The original draft of the Constitution had given Congress the power "to borrow money, and emit bills on the credit of the United States." Gouverneur Morris moved to strike the words "and emit bills." The motion passed 9–2. Madison's convention notes recorded that the majority "thought it advisable not to insert this power; thinking it probable that the omission would not disable the use of public notes so far as they would be safe and proper, and would only cut off the pretext for

a paper currency, and particularly for making the bills a tender either for public or private debts." Article I, Section 10 then explicitly forbade the states from emitting bills of credit or making anything but gold and silver coin legal tender. For Bancroft, this two-sided design—the power denied to Congress by omission, the power denied to the states by express prohibition—was the clearest possible evidence that the framers had closed the door to paper legal tender at every level of government.

Chase's career is, in Bancroft's telling, the story of that door being pried open and the constitution bleeding as a result.

6.2 The Hard-Money Democrat

Chase began his public career in the 1830s and 1840s embedded in the Jacksonian political tradition. The Jacksonian hard-money creed, which Chase shared, held that paper money not convertible to specie was a form of fraud on holders and debtors alike; that the issuance of bank notes by private institutions constituted an illegitimate privilege; and that sound money meant gold and silver coin, or notes immediately and unconditionally redeemable in them. Chase was a Free Soiler and then a Republican rather than a Democrat by the 1850s, but his monetary convictions retained their Jacksonian character: he distrusted paper, trusted specie, and consistently argued that mixing governmental and banking functions corrupted both.

As Governor of Ohio (1856–60) and then Senator, Chase never departed from these positions. When Lincoln appointed him Secretary of the Treasury in March 1861, Chase brought this monetary philosophy with him. His first instinct was to finance the Civil War through taxation and borrowing at market terms, without resort to inconvertible paper. He raised tariffs, promoted the new income tax, and placed bond issues with Jay Cooke and other financial houses. In his initial communications to Congress, Chase explicitly rejected proposals for a legal-tender paper currency.

6.3 The Legal Tender Acts and the 5-20 Bonds

By December 1861 the financial situation had become critical. Banks had suspended specie payments; the Treasury could not meet its obligations; military expenditures were running at rates that bond markets could not absorb at sustainable interest rates. Chase reversed course completely. He endorsed, designed, and lobbied vigorously

for the Legal Tender Act of February 25, 1862, which authorized the issuance of $150 million in Treasury notes—"United States Notes," quickly styled "greenbacks"—that were legal tender for all debts, public and private, but not convertible to specie. Subsequent acts in July 1862 and March 1863 authorized additional issues, bringing the total to approximately $450 million.

Alongside the greenbacks, Chase issued the so-called 5-20 bonds to finance the war: six-percent coupon bonds redeemable after five and payable within twenty years, whose authorizing statute (the Act of February 25, 1862) specified that interest would be paid in coin but was entirely silent on principal. Chase bridged this statutory gap with reputational commitment, marketing the bonds to the public—through Jay Cooke's enormous retail campaign—with the implicit understanding that principal, too, would eventually be repaid in gold. The commitment was extra-statutory: it rested on Chase's personal credibility as a hard-money man, not on any legislative guarantee. That would matter enormously later.

The arguments Chase offered for the Acts were purely fiscal and pragmatic: the government faced a choice between legal tender paper and either national bankruptcy or negotiated independence. He invoked the precedent of the Revolutionary War and argued that the power to issue legal-tender notes was inherent in sovereignty and necessary to the government's most vital function—its survival in armed conflict. He did not pretend that this was consistent with his earlier views; he argued only that emergency had foreclosed the options he would have preferred.

Bancroft does not excuse Chase for this. The Revolutionary precedent was, for Bancroft, precisely the wrong one to invoke: continental dollars were the most instructive episode in all of American monetary history, the reductio ad absurdum of paper money, and the primary inspiration for the Philadelphia framers' decision to close the door. To cite the Revolutionary experience as precedent was to cite the disease as a justification for a recurrence.

In the taxonomy of Sargent [2011], the Legal Tender Acts were a decisive move toward fiscal dominance. The government issued an inconvertible currency and declared it legal tender; the "monetary policy" embedded in this decision—the price level would be whatever was required to make $450 million in greenbacks cover real government expenditures—was entirely subordinate to fiscal necessity. Chase had, under pres-

sure of circumstances, crossed from the hard-money extreme of monetary dominance (specie standard, no fiat paper) to the fiscal-dominance extreme (inconvertible legal-tender paper issued at the government's discretion).

6.4 *Hepburn v. Griswold* (1870): Chase's First Reversal

Lincoln appointed Chase Chief Justice in December 1864. For the remainder of his life, Chase pursued a constitutional agenda centered on reversing the monetary consequences of his own Treasury policy.

The occasion came in *Hepburn v. Griswold* (75 U.S. 603), decided February 7, 1870. The case involved a debt of $11,250 contracted in 1860, before the Legal Tender Act, and tendered for payment in 1864 with greenbacks. The creditor, Mrs. Hepburn, argued that she was entitled to gold coin equivalent, since the contract antedated the Act that made paper legal tender. By a 4–3 vote (two seats were vacant), the Court held for the creditor. Chase wrote the majority opinion.

The *Hepburn* opinion rested on two grounds. First, Chase argued that compelling creditors to accept depreciated paper for pre-existing gold debts deprived them of property without due process in violation of the Fifth Amendment. Second, he argued that making greenbacks legal tender for such debts was not within the Necessary and Proper Clause: Congress could permissibly issue paper currency to finance the war, but the further step of compelling private creditors to accept it for pre-existing obligations was not "necessary" to any express power in the required sense. Crucially, Chase appealed not to express textual prohibition but to the "spirit" of the Constitution—a methodological choice that would shortly be turned against him.

The opinion conspicuously did not address whether Chase had been correct in 1862 to recommend the Acts. But the logical entailment was inescapable: if making greenbacks legal tender for pre-existing debts was unconstitutional, then Chase had committed a constitutional error as Secretary. Justice Samuel Miller's dissent noted quietly, and devastatingly, that the majority had struck down legislation that the Chief Justice himself had recommended and defended as a member of the executive branch. Miller also criticized Chase's appeal to the "spirit" of the Constitution as "abstract and intangible" reasoning that invited an unlimited judicial veto over congressional action.

6.5 *Knox v. Lee* (1871): The Court-Packing Reversal

The reversal of *Hepburn* came with stunning speed and in circumstances whose constitutional impropriety has been debated ever since. On the very day *Hepburn* was announced—February 7, 1870—President Grant nominated two new justices to fill existing vacancies: William Strong of Pennsylvania and Joseph Bradley of New Jersey. Chase believed, and said, that Grant had deliberately timed the nominations to reconstitute the Court and obtain a reversal. When the Court reached its new complement of nine, the Attorney General moved for reargument of the legal tender question. *Knox v. Lee* and *Parker v. Davis* (79 U.S. 457) were decided together on May 1, 1871, by a vote of 5–4, overruling *Hepburn*.

Justice Strong's majority opinion relied on the implied-powers doctrine of *McCulloch v. Maryland*: if Congress has the war power and the power to borrow money, and if a legal-tender paper currency is an appropriate means of executing those powers, the Acts are constitutional under the Necessary and Proper Clause. Strong's opinion included a remarkable passage—remarkable even by the standards of intra-tribunal comity—chiding Chase for having "forced" an earlier decision when the Court was divided and on the verge of receiving new appointees. As Bancroft observed, this was the majority lecturing the former majority about the indecorum of the very process by which the current majority had come to exist.

Chase dissented, joined by Justices Clifford and Field. His dissent insisted that the majority had abandoned the constitutional limitations on congressional power in favor of an open-ended theory of implied powers that left no meaningful constraint on federal action. Field's dissent was the most constitutionally principled of the three: the doctrine that whatever is not expressly forbidden may be exercised would "change the whole character of our government" from one of limited, enumerated powers to one of unlimited powers in its relations to the people.

Bancroft regarded the Court's composition question as itself corroborating his thesis. On a constitutional question of the first magnitude—whether the federal government may compel private citizens to accept depreciated paper in discharge of gold obligations—the outcome turned not on any change in the law or the facts but on the personnel of the Court. This was, in constitutional terms, not a "hard constraint" (in the sense of a textual prohibition too clear to circumvent) but a "soft constraint" that

yielded to political pressure once the political composition of the appointing authority changed. The same logic that had made Chase's reputational commitment on the 5-20 bond principal extra-statutory and therefore fragile had made the constitutional prohibition on legal tender extra-textual and therefore fragile.

6.6 *Juilliard v. Greenman* (1884): Legal Tender as Permanent Sovereignty

The third and final case swept away whatever remained of the *Hepburn* principle. In *Juilliard v. Greenman* (110 U.S. 421), decided March 3, 1884, Justice Gray's majority opinion—joined by eight of nine justices—held that the power to declare paper money legal tender was not limited to wartime emergencies but was a permanent attribute of national sovereignty. The statutory notes at issue had originally been issued during the war, then redeemed and reissued in peacetime under the Resumption and Refunding Act of 1878; no emergency justified them. That did not matter. Gray wrote:

> The power to make the notes of the government a legal tender in payment of private debts being one of the powers belonging to sovereignty in other civilized nations, and not expressly withheld from congress by the constitution; we are irresistibly impelled to the conclusion that the impressing upon the treasury notes of the United States the quality of being a legal tender in payment of private debts is an appropriate means, conducive and plainly adapted to the execution of the undoubted powers of congress.

For Bancroft, this language was "the language of revolution." The argument from what "other civilized nations" had done had no place in interpreting a constitution deliberately designed to withhold powers that every European monarchy exercised. The argument from "not expressly withheld" converted a government of enumerated powers into one of unlimited powers: if Congress may do anything not expressly forbidden, then the framers' painstaking enumeration of powers was a dead letter. Moreover, the same nine justices who joined the *Juilliard* majority had unanimously declared, just two years earlier, that "the government of the United States is one of delegated, limited, and enumerated powers," and that "every valid act of congress

must find in the constitution some warrant for its passage." Bancroft noted acidly that the *Juilliard* opinion was therefore "in flagrant antagonism" to the Court's own recent, unanimous, and repeatedly given interpretations of the constitutional structure.

The sole dissenter was Justice Field—who had been in Chase's *Hepburn* majority and in the *Knox* dissent, and who now stood entirely alone. Bancroft's rhetorical move was to insist that Field was not a minority of one but the transmitted judgment of the founding generation: standing beside him were "invincible vouchers for the rightness of his dissent"—James Wilson, Oliver Ellsworth, and William Paterson, all three appointed to the Court by Washington from among the Convention's own framers. The institutional line that the framers had drawn at Philadelphia, that Chase had tried to restore from the bench, and that Field had defended alone, had been entirely erased.

6.7 Constitutional Tragedy and Soft Constraints

Bancroft did not treat Chase's trajectory with personal relish. He read it as symptomatic of a constitutional culture that had lost its moorings rather than as individual hypocrisy. But his framing makes the theoretical point with unusual clarity. Chase's reputational commitment to gold repayment of the 5-20 bond principal and his constitutional commitment to the invalidity of legal-tender paper were both *extra-statutory*—both existed as assurances given without explicit legislative or textual authority. Both were undone by the same political logic that had originally forced Chase to accept the Legal Tender Acts as a war measure. When the political environment changed enough to matter, neither commitment was self-enforcing.

For the purposes of Sargent [2011], the arc of Chase's career encapsulates the full range of the stability–efficiency trade-off. His early Jacksonian position was an extreme form of monetary dominance: specie only, no government paper, the fiscal authority must adjust entirely to monetary reality. The Legal Tender Acts represented a wartime capitulation to fiscal dominance: the monetary instrument becomes whatever fiscal survival requires. His judicial career was an attempt to restore monetary dominance by constitutional command—to fix in law the institutional line against which executive necessity, however acute, could not encroach.

The *Hepburn–Knox–Juilliard* trilogy demonstrates that the attempt failed, and *why*

it failed. The constitutional constraint Chase sought to enforce was a soft constraint: it rested not on language expressly prohibiting the legal tender power but on the "spirit" of the Constitution as Chase interpreted it, and on the historical evidence that the Philadelphia Convention had deliberately struck out the "emit bills" clause. Bancroft thought this soft constraint should have been treated as a hard one; the Court disagreed, and the political environment— unwilling to foreclose a fiscal instrument it had used to win a civil war—did not sustain the harder reading. Institutional lines that are not politically sustainable cannot be maintained by constitutional command alone, and Chase's career is the clearest proof.

7 The Money–Credit Line: From Bond-Backing to Real Bills, 1863–1913

The reversals catalogued above all concern the boundary between the monetary and fiscal *authorities*: who controls the supply of money, and whether the fiscal authority may commandeer the monetary instrument. But there is a related and equally contested institutional line—the line between *money* and *credit*—and the half- century following Chase's National Bank Acts is a case study in Congress repeatedly redrawing it.

7.1 Bond-Backing and the Line at Government Debt

The National Bank Act of 1864 drew the money–credit line at an unusual location: United States government bonds. National Bank Notes—the dominant hand-to-hand currency of the post-Civil War era—could be issued only against specified federal bonds deposited with the Comptroller of the Currency. That institutional officer, whose primary function was verifying the bond portfolios, could certify the soundness of every dollar in circulation. The line between what counted as "money-eligible" and what counted as mere "credit" was the difference between government bonds on one side and every other asset—commercial loans, railroad securities, warehouse receipts, farm mortgages—on the other.

In the framework of Sargent [2011] this arrangement is a form of *monetary dominance*, but with a peculiar twist: the binding constraint on the money supply was not a target

set by an independent central bank but the stock of outstanding federal debt. When the government ran the large primary surpluses of the 1870s and 1880s and retired war bonds, the supply of eligible collateral contracted, and the currency supply contracted with it regardless of any commercial demand for more money. Fiscal virtue, no less than fiscal profligacy, could thus distort the monetary instrument; the two distorted it in opposite directions, but neither direction was chosen for monetary reasons. The bond-backing requirement was a hard constraint against inflationary over-issue, but it purchased that stability by tying the money supply to the government's debt-management calendar rather than to the needs of trade.

7.2 The Inelasticity Debate as Four Decades of Disagreement about the Line

Critics from the 1870s onward called the resulting currency "inelastic." The agrarian South and West needed sharply higher volumes of currency during planting and harvest seasons; the bond-backed system offered no mechanism for such expansion. Financial panics punctuated the era with depressing regularity. The political response produced two broad coalitions, each proposing to move the money–credit line in a different direction.

Greenbackers and silver inflationists wanted to dissolve the line almost entirely. If the government simply issued more fiat currency—greenbacks to be increased rather than retired, or silver to be coined at a generous ratio—there was no need for any specific asset to "back" the money at all. This was, at the limit, the same fiscal dominance that Chase had temporarily embraced during the war: the money supply would be whatever fiscal or political convenience required. William Jennings Bryan's "Cross of Gold" speech of 1896 crystallized this position in its most politically potent form.

Asset-currency reformers drew the line elsewhere. Their complaint was not too little money but the wrong kind of backing. They argued that national banks should be permitted to issue notes against a wider range of productive assets—railroad and corporate bonds, agricultural warehouse receipts, or any "good assets" on a bank's balance sheet—rather than against government bonds alone. This was a proposal to relocate the money–credit boundary from "government debt only" to "any sufficiently sound private claim." By broadening the eligibility set, they hoped, the currency sup-

ply would be free to expand with productive activity rather than with the Treasury's debt schedule.

The Aldrich–Vreeland Act of 1908, passed in the wake of the Panic of 1907, was a temporary halfway house. It allowed groups of national banks to issue emergency currency backed by "almost any securities the banks were holding," but imposed a punitive tax schedule to ensure rapid retirement once conditions normalized. More consequentially, it created the National Monetary Commission, chaired by Senator Aldrich, whose multi-volume studies of European central banking provided the intellectual scaffolding for the Federal Reserve Act of 1913.

7.3 The Real Bills Resolution and Its Costs

The Federal Reserve Act resolved the inelasticity debate by drawing the money–credit line at a new location: short-term, self-liquidating commercial paper. Section 13 of the Act confined Federal Reserve discount operations to notes and bills "arising out of actual commercial transactions" maturing within ninety days, and explicitly excluded paper drawn to finance positions in "stocks, bonds, or other investment securities." The so-called *real bills doctrine*— also called the commercial loan theory of banking, with roots in Adam Smith's *Wealth of Nations* and a long subsequent history—became the operating principle of the new institution.

The new line distinguished money from credit not by asset class but by the *purpose and duration* of the underlying transaction. A sixty-day bill drawn against a shipment of grain was eligible because the transaction was real, productive, and self-liquidating: when the grain was sold, the bill would be repaid and the money supply would contract automatically. A call loan financing stock market speculation was ineligible. In the political rhetoric of the era the line ran between Main Street and Wall Street; in the theoretical rhetoric it ran between "money backed by real production" and "money backed by finance."

The real bills criterion solved the inelasticity problem: the money supply could now expand with productive activity rather than with the stock of federal debt. But it imported a different instability, one that Henry Thornton had identified as early as 1802 in his *Paper Credit of Great Britain* [Thornton, 1802]. By accommodating every credit demand arising from commercial transactions valued at *current prices*,

the criterion set up a positive feedback from prices to money creation to prices. In a boom, prices rise; the nominal value of commercial transactions rises; the demand for discount rises; and the Fed, faithfully applying the real bills criterion, expands the money supply—adding fuel to the expansion rather than restraining it. The criterion is inherently procyclical.

The most consequential application came in 1929–1933. The Federal Reserve Board member Adolph Miller launched a "Direct Pressure" policy: any bank seeking discount window relief had to certify that it had never made speculative loans. Banks, unwilling to submit to the implied interrogation, refrained from borrowing reserves even in the face of withering deposit outflows. Instead of discounting at the window, banks failed—nearly 9,000 of them. Friedman and Schwartz [1963] held the resulting one-third contraction of the money stock largely responsible for the Great Depression. Where the bond-backed currency of the national banking era had been passively contractionary in the face of fiscal surpluses, the real bills currency proved passively contractionary in the face of banking panics. Each line carried its own characteristic failure mode.

7.4 The Theoretical Afterlife and the Sargent–Wallace Reconsideration

The real bills debate connects directly to the framework of Sargent [2011]. In a 1982 article, Sargent and Wallace [1982] showed that within overlapping-generations models the real bills prescription—unfettered private intermediation against productive assets—can be Pareto superior to the quantity-theory alternative, which imposes restrictions on intermediation in order to control the price level. This is a version of the stability–efficiency trade-off that Sargent [2011] identifies in the fiscal–monetary domain: the quantity-theory regime purchases price-level stability at the cost of constrained intermediation; the real bills regime allows efficient intermediation at the cost of weaker price-level control. Laidler [1984] objected that the Sargent–Wallace version had abstracted away the key feature of the historical doctrine—the restriction to specific collateral types—but the theorem demonstrated that the money–credit distinction is not merely an institutional accident or a historical curiosity. It reflects a genuine analytical trade-off whose terms depend on the structure of the economy and the quality of the regulatory environment.

The congressional progression from bond-backed currency to real bills—from the National Bank Acts of 1863–64 to the Federal Reserve Act of 1913, through four decades of inelasticity complaint and asset-currency agitation—thus illustrates in miniature the same dynamic as the individual reversals catalogued in this paper. The actors who drafted the Federal Reserve Act were consciously reversing the institutional design that Chase had supervised as Secretary of the Treasury. Their reversal solved one class of problems while creating another; and the real bills doctrine embedded in the new Act would itself require substantial revision in the Banking Acts of 1932–35, a revision precipitated by the very failure mode that Thornton had foreseen in 1802. The half-century arc from National Bank Notes to Federal Reserve notes is another turn of the same wheel: each new institutional line is drawn in response to the failures of the previous one, and each generates its own successor crisis.

7.5 Peel's Bank Charter Act and Its Three Suspensions: A British Parallel

The dynamic of statutory lines tested and bent by crisis has a precise British parallel in the Bank Charter Act of 1844. Robert Peel's Act divided the Bank of England into two departments: an Issue Department, which could issue notes only against gold or a fixed fiduciary limit of £14 million; and a Banking Department, which could lend freely but had no special link to note issuance. The Currency School theorists who designed this structure—Samuel Jones Loyd (Lord Overstone) was their foremost theorist—intended it as the hardest possible statutory constraint on money creation [Fetter, 1965]. By tying the note issue to gold and banning discretionary accommodation, they believed they had drawn the line against inflationary over-issue once and for all.

The Act was suspended by Treasury letter in October 1847, November 1857, and May 1866—in every major financial panic within a generation of its passage. In each case the mechanism was the same: the Bank of England, constrained by the Act from expanding its note issue, faced a choice between watching the banking system collapse and requesting from the government an indemnity letter authorizing temporary breach of the statutory limit. The government wrote. In 1847, the announcement alone proved sufficient to calm the panic; the Bank's note issue never in fact breached the legal maximum. In 1857, following the failure of the Western Bank of Scotland

and the Glasgow City Bank, the Bank did exceed its fiduciary limit and Parliament subsequently provided indemnity. In 1866, after the spectacular collapse of Overend, Gurney and Company—then the largest bill broker in London—the Bank initially hesitated to lend freely, exacerbating the panic before the suspension letter arrived.

Walter Bagehot's response in *Lombard Street* [Bagehot, 1873] constitutes the most intellectually consequential commentary on this pattern. Bagehot was not an opponent of the Bank Charter Act on Currency School grounds; he accepted its basic institutional structure as too politically entrenched to repeal. But his prescription inverted Peel's design. Where Peel's Currency School architects had intended the gold constraint to force contraction in a crisis—compelling banks to liquidate and prices to fall, rather than accommodating speculative excess—Bagehot argued that in a genuine panic the Bank *must* lend freely at a penalty rate against good collateral. The lender-of- last-resort function that *Lombard Street* articulated was the theoretical mirror image of everything the Currency School had built the Act to prevent.

In the terms of Sargent [2011], the Bank Charter Act was an attempt to inscribe monetary dominance into statute: the note issue would be bound, the banking or fiscal residual would adjust. The three suspensions demonstrated that the attempt produced a statutory line rather than an institutional reality. The line was not changed; it was, as Bagehot saw, managed by the publicly known expectation that it would be suspended whenever truly tested. "The practical consequence is that the Bank of England is obliged to keep the reserve for the whole banking system," Bagehot wrote, "and yet no one has said it should; and no one will say it should." The constraint existed on paper; the practice had already moved past it.

The parallel to the *Hepburn–Knox–Juilliard* trilogy is exact. In both cases a monetary rule was established by law or constitutional convention to prevent fiscal or speculative accommodation. In both cases the rule was suspended under pressure—by executive letter in Britain, by court-packing in America. In both cases the exception tended to become the norm: repeated Bank Act suspensions made suspension a publicly anticipated contingency, just as *Juilliard* made legal tender a permanent attribute of sovereignty. And in both cases the most penetrating analytical response was not a defense of the original line but a new doctrine that institutionalized the exception: Bagehot created the lender-of-last-resort doctrine to rationalize the accommodation; Bancroft could only mourn that the Philadelphia framers' judgment had been overrid-

den. The difference in outcome reflects a difference in institutional creativity: Britain produced a reformulation that preserved the spirit of stability while accepting the necessity of crisis lending; America consumed the original constraint without producing a comparably coherent successor.

8 Twentieth-Century Reversals

The individual and institutional reversals catalogued above all belong, in the main, to the nineteenth century and the first years of the twentieth. Three twentieth-century episodes extend the same pattern with additional precision, and together they illuminate dimensions of the stability–efficiency trade-off that the earlier record leaves implicit.

8.1 John Maynard Keynes: From Gold Standard Defender to Its Foremost Critic

Among economists, the most consequential reversal of the twentieth century belongs to John Maynard Keynes. Its direction is the mirror image of Friedman's: where Friedman moved from a design coupling monetary policy to fiscal outcomes toward a strict monetary rule independent of the budget, Keynes moved from sophisticated defense of a specie-based monetary constraint toward the most influential advocacy of managed money and fiscal flexibility in the English language.

Keynes's starting point was *Indian Currency and Finance* [Keynes, 1913], published in the same year as the Federal Reserve Act. The book analyzed India's gold exchange standard with broadly affirmative conclusions: the system was, Keynes argued, a rational institutional evolution that captured the price-stability benefits of gold convertibility without requiring the costly maintenance of a full gold circulation. He commended the system's efficiency and stability. The book was received—and later remembered, not entirely fairly—as evidence of a youthful attachment to specie discipline [Moggridge, 1992].

Within a decade the world that book described had been destroyed by the Great War. Britain had suspended gold convertibility in 1914; the United States had accumulated the bulk of the world's monetary gold; wartime inflation had restructured

debt relationships in ways that made the restoration of pre-war parities a program of systematic deflation rather than of genuine price stability. By 1923, when Keynes published *A Tract on Monetary Reform* [Keynes, 1923], his position had inverted. The gold standard, he wrote, was "already a barbarous relic"; the sensible objective was to manage the price level directly through central bank discretion rather than to subordinate domestic economic conditions to the requirements of an external specie constraint. The book proposed a "managed currency" in which the central bank would target some combination of price-level and output stability, using its instruments actively rather than following a specie rule. This is, in the taxonomy of Sargent [2011], a shift from monetary dominance (the gold standard binds the monetary authority, forcing fiscal and wage adjustment) toward managed discretion—not quite fiscal dominance, but a regime in which the monetary authority responds to economic conditions rather than to a fixed specie constraint.

The intellectual break became concrete in "The Economic Consequences of Mr. Churchill" [Keynes, 1931], originally published in 1925. Keynes attacked Winston Churchill's decision to return Britain to the gold standard at the pre-war parity of $4.86. The pre-war rate overvalued sterling by approximately 10 percent relative to the dollar, requiring British wages and prices to fall by that margin to restore international competitiveness. Keynes predicted with precision the consequences that followed: industrial contraction, the coal dispute that became the General Strike of 1926, prolonged unemployment in the export industries, and a deflationary drag that persisted until Britain finally abandoned gold in 1931. The episode confirmed for Keynes—and for many who watched—that the price of monetary discipline under gold could be measured not in abstract stabilization calculus but in unemployment rates and broken industries.

Keynes's reversal illuminates the logic of Sargent [2011] precisely because it mirrors Friedman's in structure while inverting its conclusion. Both were updating the same trade-off in response to the same cluster of historical events—the collapse of the pre-war monetary order, the Great War, the interwar deflation—but each concluded that the other's preferred failure mode was the more serious danger. Keynes, observing the interwar years from the vantage of deflationary Britain, concluded that the cost of adjustment under specie discipline was the dominant risk: the monetary authority must retain discretion to prevent deflation, even at the cost of a weaker price-level

anchor. Friedman, composing the *Monetary History* with Schwartz [Friedman and Schwartz, 1963], concluded that the failure of the monetary authority to maintain a stable money supply under discretionary management was the dominant risk: the central bank must be bound by a rule, even at the cost of fiscal inflexibility. Each was correct about the failure mode he emphasized, and each was less attentive to the failure mode emphasized by the other. Eichengreen [1992] shows at length how the gold standard constrained monetary responses to the Depression; Friedman and Schwartz [1963] demonstrates how the Federal Reserve's discretionary failures deepened it. Both accounts are accurate. The divergent conclusions that Keynes and Friedman drew from the same evidence are not a sign that one was wrong; they are the analytical content of the no-dominant-option result of Sargent [2011].

8.2 Marriner Eccles and the Treasury–Fed Accord of 1951

The twentieth century's most structurally precise parallel to Chase's arc belongs to Marriner Eccles. As Chairman of the Federal Reserve from 1934 to 1948, Eccles championed the wartime arrangement under which the Fed pegged Treasury bond prices—holding 90-day bill yields at $\frac{3}{8}$ percent and long bond yields at $2\frac{1}{2}$ percent regardless of monetary conditions. Like Chase's endorsement of the Legal Tender Acts, this was a deliberate crossing to fiscal dominance under wartime pressure: the monetary authority became an unlimited buyer of government securities at whatever volume was needed to keep the government's borrowing costs low. Seigniorage was whatever the wartime fiscal program required; the price level was the fiscal residual. Eccles understood the inflationary risk and accepted it as the price of war finance. He said so explicitly in congressional testimony and in his memoir [Eccles, 1951].

The parallel to Chase deepens in the postwar period. As wartime controls were lifted and inflation accelerated, the bond-price peg required the Fed actively to expand its balance sheet to prevent bond yields from rising. This meant accommodating inflationary pressure rather than resisting it—a role that Eccles, who had championed the peg for wartime reasons, now found untenable. By 1948, his public and internal arguments for restoring Federal Reserve independence had become a political liability. President Truman demoted Eccles from the chairmanship to an ordinary Board seat, replacing him first with Thomas McCabe. Like Chase, who used his Chief Justiceship to attack the Acts he had himself designed, Eccles used his reduced institutional po-

sition to keep pressing for monetary independence—and, like Chase, he was penalized for the attempt. The man who had designed and championed the fiscal-dominance arrangement was removed from its administration precisely because he had concluded it needed to end.

The Treasury–Fed Accord of March 4, 1951, negotiated by McCabe's successor William Martin, formally ended the peg [Hetzel and Leach, 2001]. The Fed regained the authority to set interest rates to serve monetary objectives without regard to Treasury borrowing costs; monetary dominance was restored. The Accord became, in subsequent decades, the founding charter of American central bank independence—the institutional line whose integrity each successive episode of fiscal pressure would test [Meltzer, 2003].

What explains the different outcome compared to Chase's trajectory? Where Chase's *Hepburn* decision was reversed by *Knox* and extinguished by *Juilliard*, the Accord held. Several features of the postwar environment appear decisive. First, the inflationary consequences of the peg were visible and politically salient in a way that the deflationary consequences of Civil War-era monetary contraction were not. The peg was producing inflation that hurt middle-class savers; the political coalition for monetary independence could therefore extend beyond financial elites. Second, the Korean War, while generating some pressure for continued Treasury accommodation, did not produce the existential fiscal emergency of 1861–65; borrowing requirements were manageable without monetization. Third—and perhaps most importantly—the Fed had a credible and immediately exercisable alternative: raise interest rates today, without any legislative action. Chase, by contrast, needed the Court to strike down a statute; the institutional inertia ran in the opposite direction. Soft constraints succeed when the political environment can sustain them; the Accord held because the inflation tax imposed by the peg had created a constituency for its removal that the Civil War greenbacks had not.

8.3 Richard Nixon and the End of Bretton Woods

If Chase's career is the most internally dramatic of the nineteenth-century reversals, Richard Nixon's New Economic Policy of August 1971 is its twentieth-century counterpart in terms of surface paradox. Nixon had built his political identity as the anti-Keynesian: his 1968 campaign attacked Lyndon Johnson's "guns and butter" policy,

the inflationary combination of Vietnam War spending and Great Society expansion without compensating tax increases. He opposed discretionary deficit spending and price controls, the signature instruments of Keynesian demand management. His initial Council of Economic Advisers chairman, Paul McCracken, was a monetarist-inflected fiscal conservative.

The Bretton Woods system—linking the dollar to gold at $35 per ounce and other currencies to the dollar—was the twentieth century's international analog of the gold standard: a form of monetary dominance at the systemic level, in which the dollar's gold peg constrained American fiscal and monetary policy in precisely the way the domestic gold standard had constrained individual economies in Keynes's time. By 1971, years of persistent balance-of-payments deficits, generated by the fiscal costs of Vietnam and the Great Society, had depleted the American gold stock and filled foreign central banks with dollars they were increasingly inclined to convert. The fiscal dominance latent in Johnson's spending decisions had gradually undermined the monetary constraint, just as Sargent–Wallace "unpleasant arithmetic" would predict.

On the evening of Sunday, August 15, 1971, Nixon announced the New Economic Policy. In a single televised address he simultaneously suspended dollar–gold convertibility (ending Bretton Woods), imposed a ninety-day freeze on wages and prices (the instrument he had specifically opposed), and announced a Keynesian stimulus package designed to ensure re-election in 1972 [Matusow, 1998]. He told his economic advisers, in the phrase that became the shorthand for the entire episode, "I am now a Keynesian in economics."

Nixon's reversal differs from the wartime reversals in one important respect. It was not compelled by active military emergency in any proximate sense. Vietnam was ongoing, but the balance-of-payments pressure that broke Bretton Woods was fundamentally fiscal: the accumulated legacy of years of "guns and butter" spending, which had inflated the dollar monetary supply relative to gold reserves. The connection to war was mediated through fiscal consequences rather than through the direct demands of military finance. This is the peacetime version of the mechanism Sargent [2011] identifies: when fiscal dominance is the revealed preference of the political system, no external monetary anchor can hold indefinitely against the accumulated pressure, even in the absence of immediate armed conflict.

The temporary also became permanent, as it had after the Civil War. Nixon announced the gold window closure as provisional, pending international monetary reform. Reform came, in qualified fashion, at the Smithsonian Agreement in December 1971 and at Kingston in 1976; but the dollar's gold link was never restored. Gowa [1983] documents the political economy of this irreversibility in detail, showing that the domestic fiscal commitments which had caused the balance-of-payments crisis were not retracted, and without their retraction the gold price could not have been defended. The emergency of August 1971, like Chase's wartime greenbacks, became the permanent monetary constitution of the world it had transformed.

9 Common Themes

Several themes recur across these reversals.

War and systemic crisis as forcing functions—and the clustering of reversals. The reversals documented here cluster at moments of acute fiscal and financial stress. The War of 1812 converted Clay and Madison simultaneously, and vindicated Gallatin's earlier administrative conclusions. The Civil War broke Chase, constructed the bond-backed currency system, and established the fiscal-dominance precedent that *Juilliard* would later entrench as permanent. Britain's participation in the Great War destroyed the international gold standard that Keynes had analyzed affirmatively in 1913 and set in motion the deflationary chain that drove his 1923 reversal. World War II created the wartime peg that placed Eccles at the fiscal-dominance extreme; postwar inflation drove him to the opposite pole. Nixon's 1971 reversal was the delayed fiscal consequence of Vietnam spending accumulated over Eisenhower, Kennedy, and Johnson administrations.

This clustering is not coincidental. In a steady fiscal environment, the costs of the existing institutional arrangement—however imperfect—are broadly distributed across time and may be tolerable. Under the concentrated demands of war or systemic panic, they become immediate and undeniable. The fiscal-dominance arrangement—using monetary instruments to finance emergency expenditures—becomes irresistible precisely because the alternative is defeat or systemic collapse. This is not an argument against institutional constraints on fiscal-monetary coordination; it is a reminder that any such constraints must be evaluated in part by their behavior under the

extreme stress cases that history periodically and unpredictably generates. Sargent and Wallace [1981] and Sargent [2011] focus on steady-state comparisons of alternative institutional regimes. The historical record adds a dynamic dimension: the transitions between regimes—forced by crisis—may matter as much as the steady states themselves. A well- designed monetary-dominant regime that collapses under wartime pressure may leave the fiscal authority worse off than a more flexible fiscal-dominant regime that never pretended to prohibit seigniorage. The value of an institutional commitment depends not only on its steady-state properties but on its resilience when tested.

The reflexivity of experience. Each of the reversals was accompanied by an appeal to experience that the original position could not have anticipated. Clay and Madison could not have known in 1791 or 1811 what the War of 1812 would reveal about the fiscal consequences of the absence of a national monetary institution. Friedman could not have known in 1948 what his and Schwartz's subsequent historical research would reveal about the monetary origins of the Great Depression, or what the postwar Keynesian political economy would reveal about the systematic bias toward fiscal expansion. Keynes could not have known in 1913 what the Great War's fiscal demands would do to the monetary system whose workings he had so carefully described. Experience updated the effective parameters of the trade-off identified in Sargent [2011]: the relative magnitude of the fiscal and monetary risks, and the capacity of political institutions to manage them.

The constitutional versus the pragmatic argument. Chase's career and Madison's liquidation doctrine both reveal a tension between constitutional arguments about where institutional lines may be drawn and pragmatic arguments about where they should be drawn. The constitutional argument tries to fix the line against the pressures of the moment; the pragmatic argument allows the line to respond to changing circumstances. Madison resolved the tension by treating constitutional meaning as itself partly determined by experience (the liquidation doctrine). Chase resolved it by insisting that the Constitution fixed the line absolutely—and discovered that the political economy would not sustain that absolutism. Peel's Bank Charter Act illustrates the same tension in statutory form: a rule rigid enough to command credibility was not flexible enough to survive a genuine crisis, and Bagehot's response—accept the constraint on paper while institutionalizing its suspension as lender-of-last-resort

doctrine—was the most intellectually honest resolution available.

The ratchet problem. Friedman's final position—the fixed money-growth rule—was motivated in part by his assessment that the coupling of monetary and fiscal policy in a democracy reliably moves in one direction: toward more money creation, more inflation, more fiscal expansion. The historical record broadly supports this concern. Neither Clay nor Madison, having supported the Second Bank, sought to use it as an instrument of fiscal dominance; but Jackson's destruction of the Bank was followed by the pet-bank era, the suspension of 1837, and eventually the Legal Tender Acts. Nixon announced the gold window closure as temporary; it became permanent. Chase's emergency greenbacks were validated by *Juilliard* as a permanent attribute of sovereignty. The ratchet runs toward fiscal dominance when monetary institutions are weak; only durable, credible monetary commitments can reverse it, and such commitments are—as Eccles and Volcker both discovered—fragile in the face of sustained fiscal pressure.

The impossibility of personal commitment. Perhaps the deepest lesson of Chase's career is that individuals cannot, through the force of their own convictions, maintain institutional lines against political and fiscal pressures. Chase's personal monetary principles were genuine and deeply held; he had stated them publicly for decades. Under sufficient pressure—the fiscal demands of a war for national survival—he abandoned them. And having abandoned them, he could not restore them from his judicial position, because the political forces that had driven the abandonment had not themselves changed. Eccles faced the same discovery: he could advocate for monetary independence from a demoted Board seat, but he could not restore it unilaterally. Institutional lines must be maintained by institutions, not by individuals; and as Madison's liquidation doctrine suggests, institutions themselves are mutable.

The asymmetry of conclusions from the same evidence. The most intellectually arresting observation to emerge from placing the reversals side by side is that the same historical events can drive different analysts to opposite conclusions. The interwar monetary disorder drove Keynes toward managed money and fiscal flexibility and drove Friedman toward a binding monetary rule. The Great Depression confirmed for Keynes that deflation under gold discipline was unsustainable; for Friedman, it confirmed that monetary contraction under discretionary management was the primary danger. Both conclusions were correct about the failure mode each emphasized;

each was less attentive to the failure mode emphasized by the other. This is not relativism: it is the analytical content of the no-dominant-option result of Sargent [2011]. The correct institutional conclusion from the same historical experience depends on a judgment about which of two genuine risks is the greater one—and that judgment, as the cases show, can differ between equally careful observers reasoning from identical evidence.

10 Conclusion

The trade-off between price-level stability and Ramsey-optimal fiscal efficiency identified in Sargent [2011] is not merely a property of theoretical models. It is a live institutional choice that has generated passionate disagreement, principled reversal, and constitutional crisis throughout American and British economic history. The nine individual cases examined here—Friedman's shift from fiscal coordination to monetary rules; Volcker's adoption and abandonment of Friedman's rule; Clay's conversion from Republican opponent to Whig champion of the national bank; Webster's progression from opposing the Second Bank's charter to serving as its legal champion; Gallatin's transformation from Jeffersonian critic of Hamilton's Bank to its strongest executive-branch advocate; Madison's move from strict-constructionist critic to pragmatic signer of the Second Bank bill; Chase's extraordinary journey from hard-money Democrat to greenback architect to constitutional counter- revolutionist; Keynes's passage from gold-exchange defender to the author of "barbarous relic"; and Eccles's arc from wartime fiscal-dominance architect to champion of the Accord that restored monetary independence—all illustrate the same underlying truth: the optimal institutional line responds to circumstances, and circumstances change. Nixon's New Economic Policy adds a further illustration: when fiscal dominance is the revealed preference of the political system, no external monetary anchor can hold indefinitely, even in peacetime.

That truth is confirmed, at the legislative level, by the half-century redrawing of the money–credit line examined above. The bond-backed currency of the National Banking System was a stable but inflexible answer to the question "what should money be backed by?" The real bills doctrine of the Federal Reserve Act offered a more elastic answer that solved the inelasticity problem while importing the procyclical

instability that Thornton had identified in 1802 and that Friedman and Schwartz held responsible for the catastrophe of 1929–33. Peel's Bank Charter Act across the Atlantic tells a parallel story: a monetary rule inscribed in statute proved unable to survive three successive crises without executive suspension, and the most penetrating response was not a repeal of the rule but Bagehot's reformulation—transforming the exception into a doctrine of active crisis lending. The theoretical reconsideration by Sargent and Wallace [1982], showing that a real-bills regime can be Pareto superior to a quantity-theory alternative, demonstrates that the congressional reversal from bond-backing to real bills, like the individual reversals, reflects a genuine analytical trade-off rather than mere confusion.

This is not an argument for institutional nihilism. The historical record also supports Friedman's conviction that price-level stability requires commitments that are robust to fiscal pressure, and that such robustness is hard to achieve and easy to destroy. The Second Bank was chartered in 1816 in response to the fiscal lessons of 1812 and destroyed by 1836 in response to political pressures that its designers had not foreseen. The Legal Tender Acts were upheld by a reconstituted Court and confirmed as permanent sovereignty in *Juilliard*. The Bretton Woods gold link, announced as temporary in August 1971, was never restored. The trajectory is one of progressive erosion of monetary constraints under fiscal pressure—what the paper has called the ratchet.

The cases do counsel, however, for humility about institutional design. Principled people who have thought carefully about where to draw the lines have repeatedly found themselves unable to maintain those lines in the face of changed circumstances, new evidence, and the coercive logic of fiscal emergency. That reversals cluster at historical stress points—wars, panics, depressions—is not surprising; it is the analytical content of Sargent and Wallace [1981]. What is surprising, and sobering, is that the reversals are so rarely provisional: the emergency measure typically becomes the permanent settlement. Designing institutions that are both responsive enough to survive systemic stress and durable enough to provide the stability that is their purpose is the enduring challenge. The reversals documented here are a measure both of how difficult that challenge is, and of how much depends on getting the answer right.

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